

CORNERSTONE Off-the-Shelf Demonstration Chip Guidelines

I. Chip overview

The off-the-shelf demonstration chips were fabricated on the standard CORNERSTONE 220 nm silicon-on-insulator (SOI) platform with a 1 μm SiO_2 top cladding and a 2 μm buried oxide (BOX) layer. Metal heaters were integrated as thermo-optic phase shifters. Most of the devices work at 1550 nm wavelength and if it is not specified, the working wavelength is 1550 nm when these devices are discussed.

The layout of this demonstration cell is shown in Fig. 1.

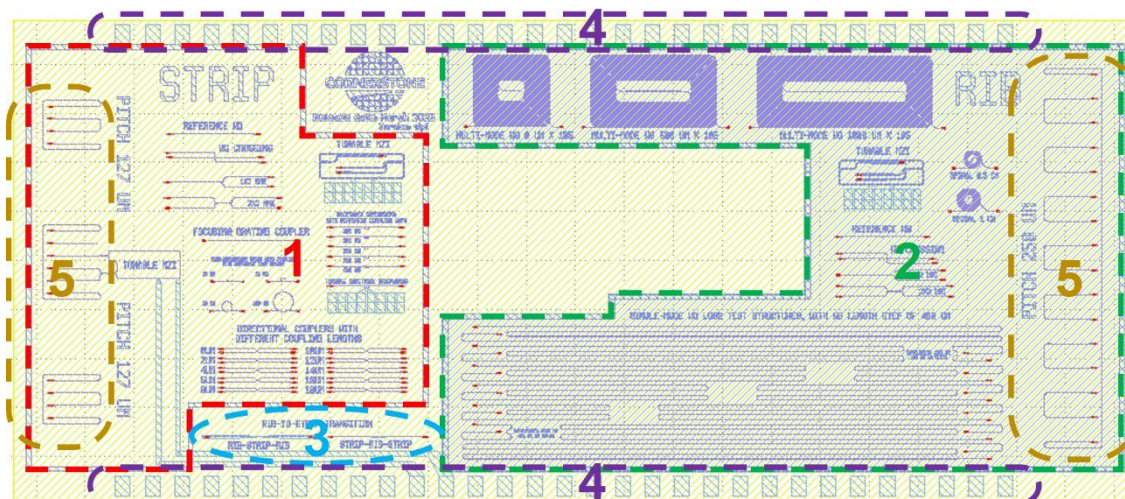


Figure 1. Layout overview.

Generally, there are five groups of devices/structures and they are:

1. Photonic devices based on strip waveguide
2. Photonic devices based on rib waveguide
3. Rib-to-strip transitions
4. Electrical contact pads for wire bonding
5. Grating coupler arrays for optical fibre bonding

Each component is labelled physically on the chip, which is visible under a microscope. A blank area in the center of the PIC chip is reserved for customer designs.

II. Photonic devices based on strip waveguides

1. Basic components

The most basic components in a photonic integrated circuit (PIC) are straight waveguides, waveguide bends, waveguide crossings, and beam splitters/couplers.

As shown in Fig. 2, there are four devices circled, (1) a short straight waveguide in between two sets of grating coupler and taper for light input and output, (2) a waveguide crossing with four input/output grating couplers, (3) a 1x2 multi-mode interferometer (MMI) beam splitter/coupler with input/output grating couplers, and (4) a 2x2 MMI beam splitter/coupler with input/output grating couplers.

In Device (1), the optical loss of the short straight waveguide is negligible and the transmission of this device reflects the transmission of the two grating couplers. A peak transmission of -11 to -15 dB is expected.

In Device (2), for example, when the light is launched into the device from Port 1, a peak transmission of -11 to -15 dB is expected from Port 4 because the loss of the crossing is insignificant, whilst the transmission from Port 2 and Port 3 should be negligible.

In Device (3), assuming the light is launched into the device from the left port, similar spectra are expected from the two ports on the right, which means the 1x2 MMI can evenly split the optical power.

In Device (4), which is similar to (3), assuming the light is launched into the device from one of the two left ports, similar spectra are expected from the two ports on the right, which means the 2x2 MMI can also evenly split the optical power. Please note: it is difficult to realize a exactly even splitting using a 2x2 MMI especially in a wide wavelength range. Also there is $\pi/2$ phase difference on the output ports.

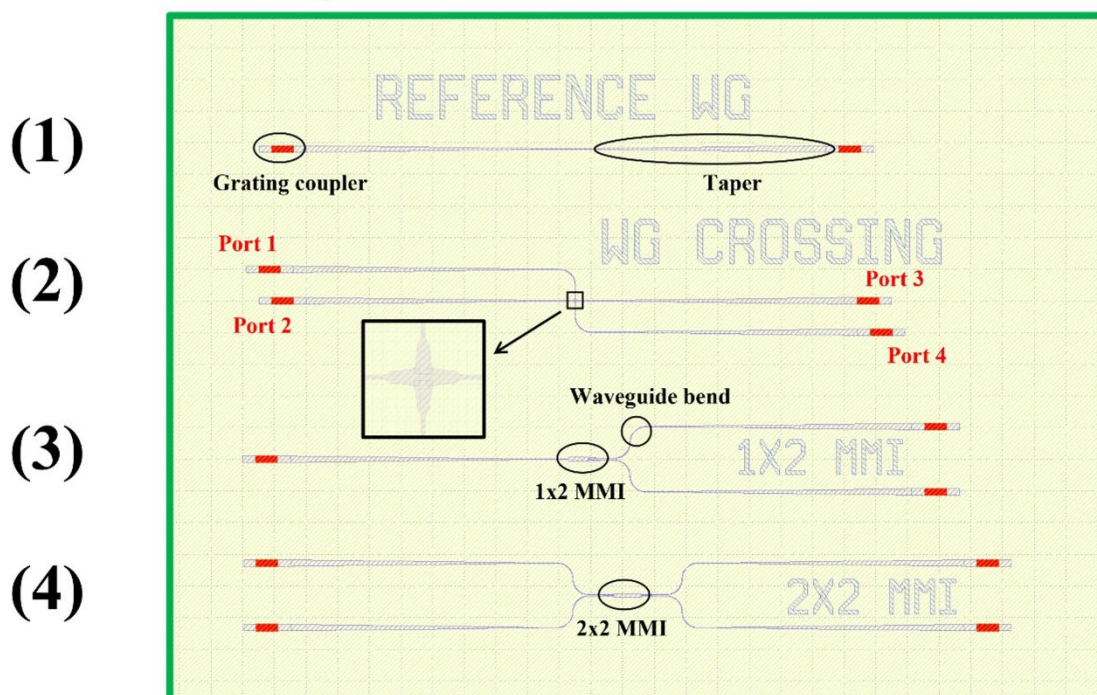
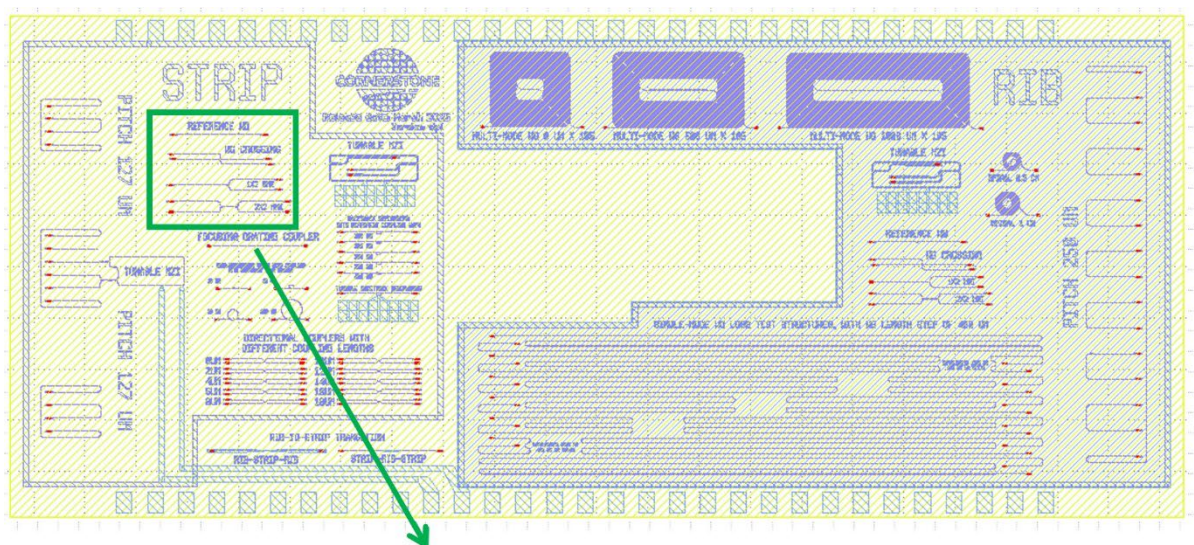


Figure 2. Basic components.

Please note that the straight waveguides in this section and following sections (if not specified) have a width of 450 nm and they are single mode (SM) transverse-electric (TE) waveguides. The waveguide bends also have a width of 450 nm. All the bends used on this demonstration chip are arc bends and thus it has to support single mode only.

2. Tunable MZI

Using two MMIs one can easily build a Mach–Zehnder interferometer (MZI). See the device in Fig. 3 as an example of a MZI, which is formed by a 1x2 MMI, 2x2 MMI, straight waveguides, waveguide bends and grating couplers. On the transmission spectrum, one should observe periodic constructive and destructive interference. The period is the so-called free-spectral-range (FSR), which is correlated with the optical path difference of the two MZI arms between the 1x2 MMI and 2x2 MMI.

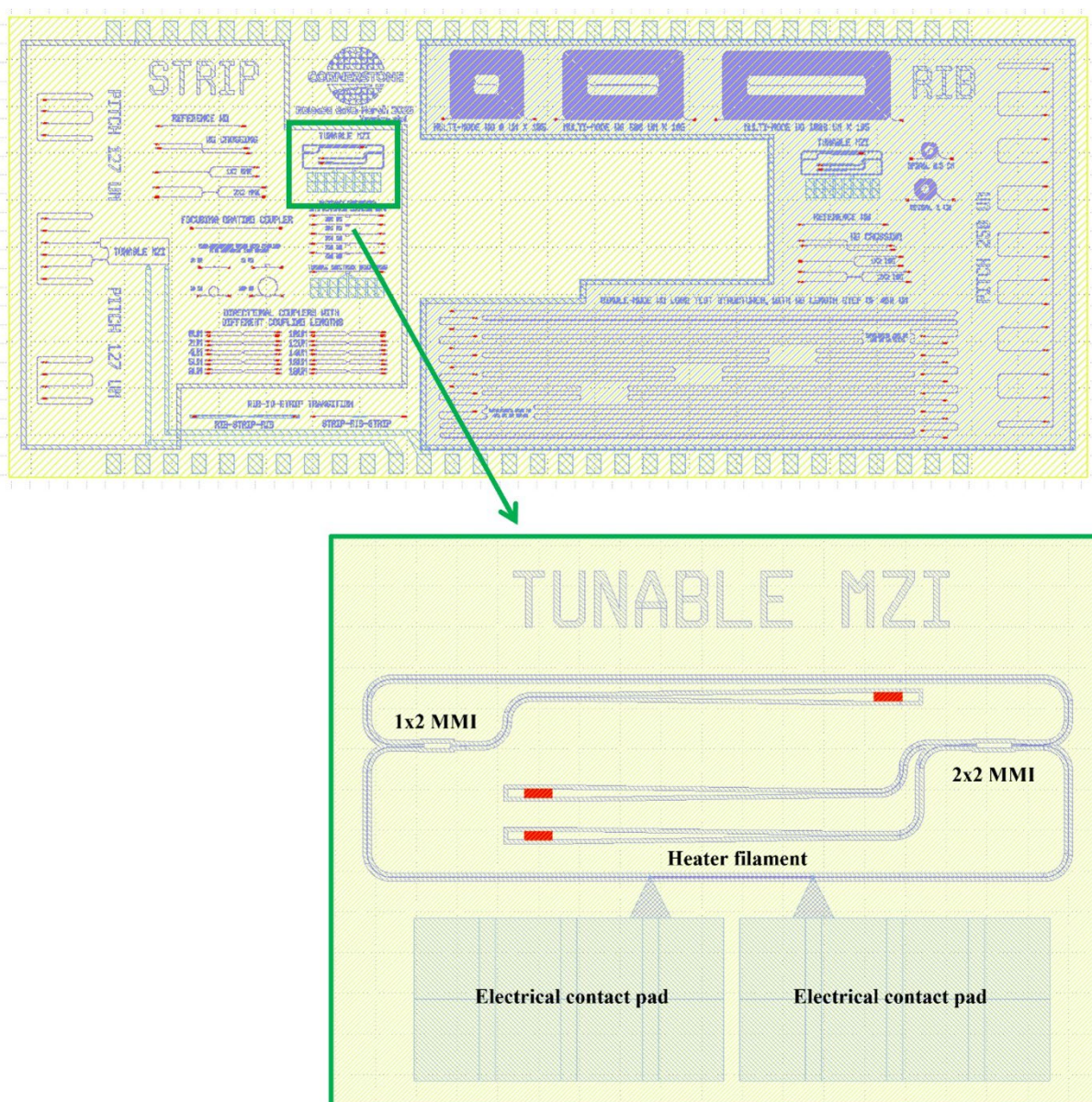


Figure 3. Tunable MZI.

Please note that a thermo-optic phase shifter is integrated on one of the MZI arms. When a voltage difference is applied to the two contact pads, there will be a current flow through the filament to heat the MZI arm up, eventually resulting in the effective arm length difference change. Thus when different voltages are applied,

the constructive/destructive wavelength and the FSR can be tuned. Or, if the input wavelength is fixed and the output optical power is monitored, when the heating power is linearly increased, the optical power should vary as a sinusoidal curve. Please note these contact pads are designed for probes, whilst they can be bonded as well.

3. Directional couplers

As shown in Fig. 4, there are a series of directional couplers with various coupling lengths. When two waveguides are placed in parallel and close enough with each other, they are coupled evanescently and the optical power will exchange between them. With a fixed coupling gap, the power exchange rate (power per unit distance) is fixed. By varying the coupling length, different amounts of optical power will be transferred from one waveguide to the other. So, if the light is launched into the device from one of the grating couplers, the transmission ratio at the grating couplers on the other side will be different with different coupling length.

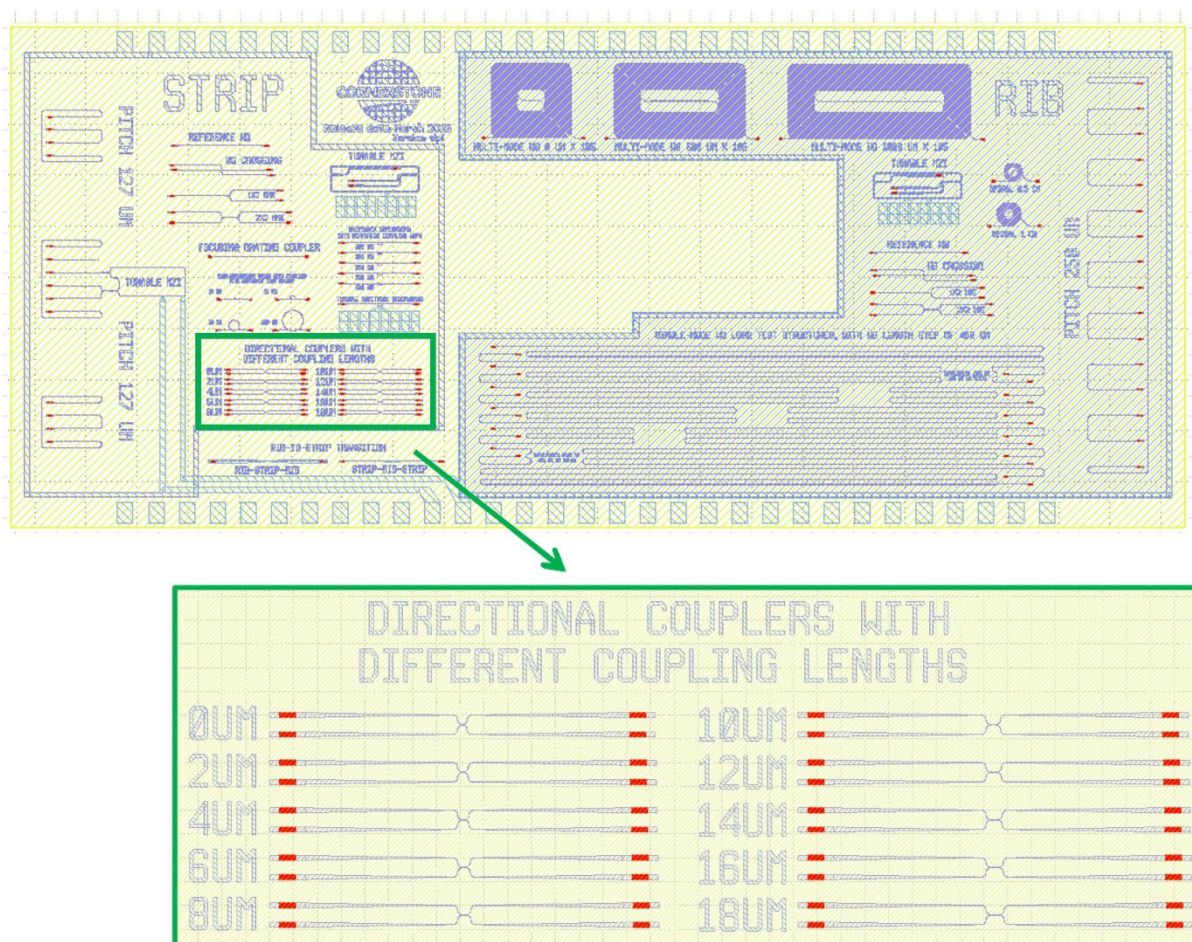
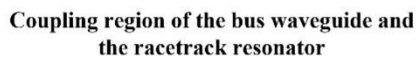


Figure 4. Directional couplers.

The directional coupler is also widely used in PICs. For example, it is the key part of a racetrack resonator.

4. Racetrack resonators

Ring/racetrack resonators are the most common resonant components used in PICs. Some examples of racetrack resonators are shown in Fig. 5. The top four racetrack resonators are fixed components, with different coupling gaps. Periodic resonance dips, which denote constructive interference in the resonator are expected in the transmission spectrum. The period, that is FSR, is mainly determined by the circumference of the racetrack, whilst the coupling gap has slight effect on the FSR. On the other hand, the coupling gap is a key factor in the design and it controls the Q factor and the coupling depth (extinction ratio) of the resonances.



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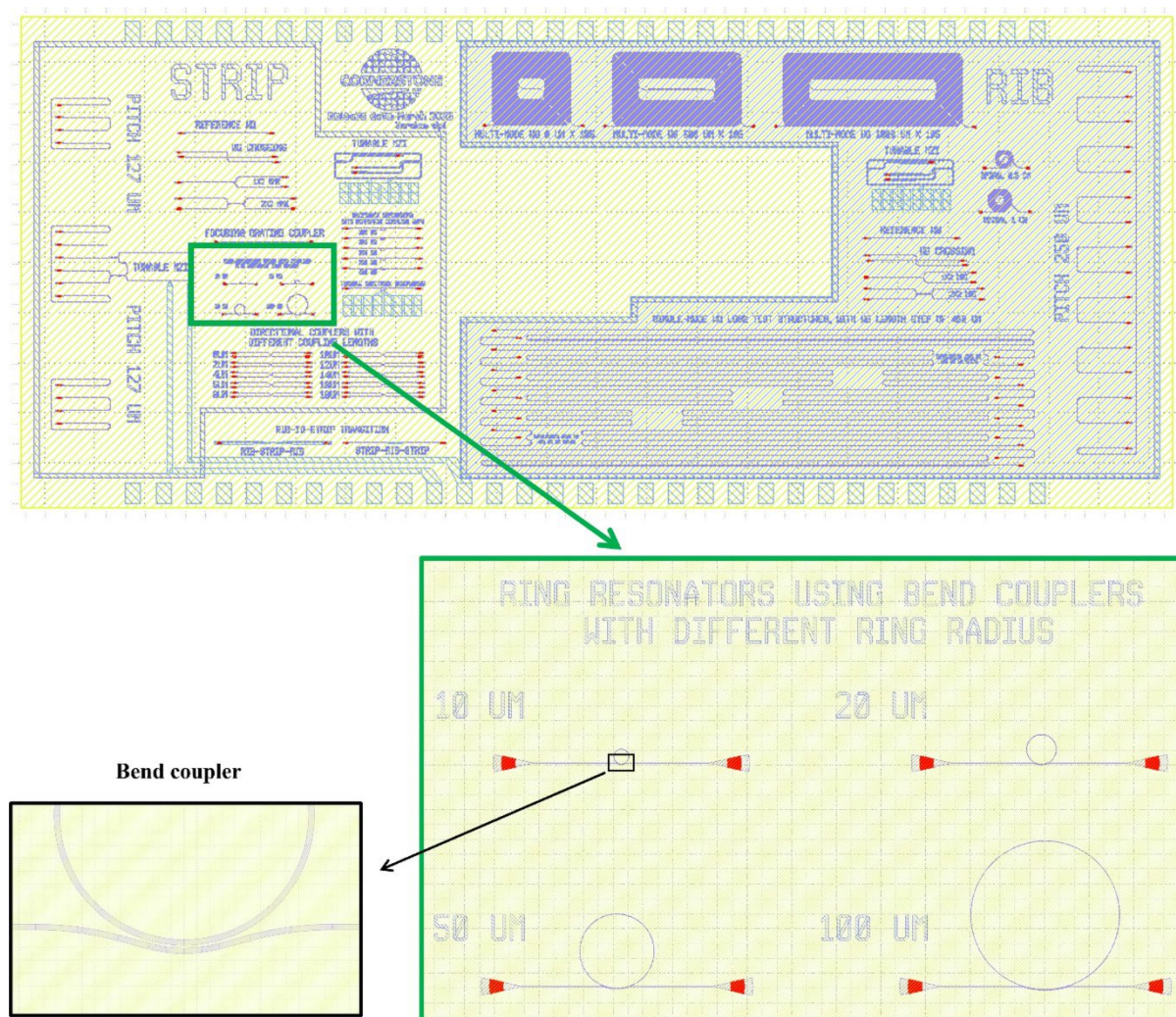


Figure 6. Ring resonators.

6. Focusing grating coupler

One must note that the grating coupler used in the ring resonators are different from before. The grating couplers discussed above in Chapter II, Section 5 are all regular gratings, in which the grating teeth are straight. This kind of design requires a long taper to connect it to a single mode waveguide, which is very space consuming. To conquer this problem, one can curve the grating teeth, as shown in Fig. 7, to form the so-called focusing grating coupler.

Compared with regular grating design, this kind of focusing design can save a lot of space. However, the design is less robust and the fibre alignment to this kind of grating requires a tighter tolerance. The transmission performance should be similar to the regular one.

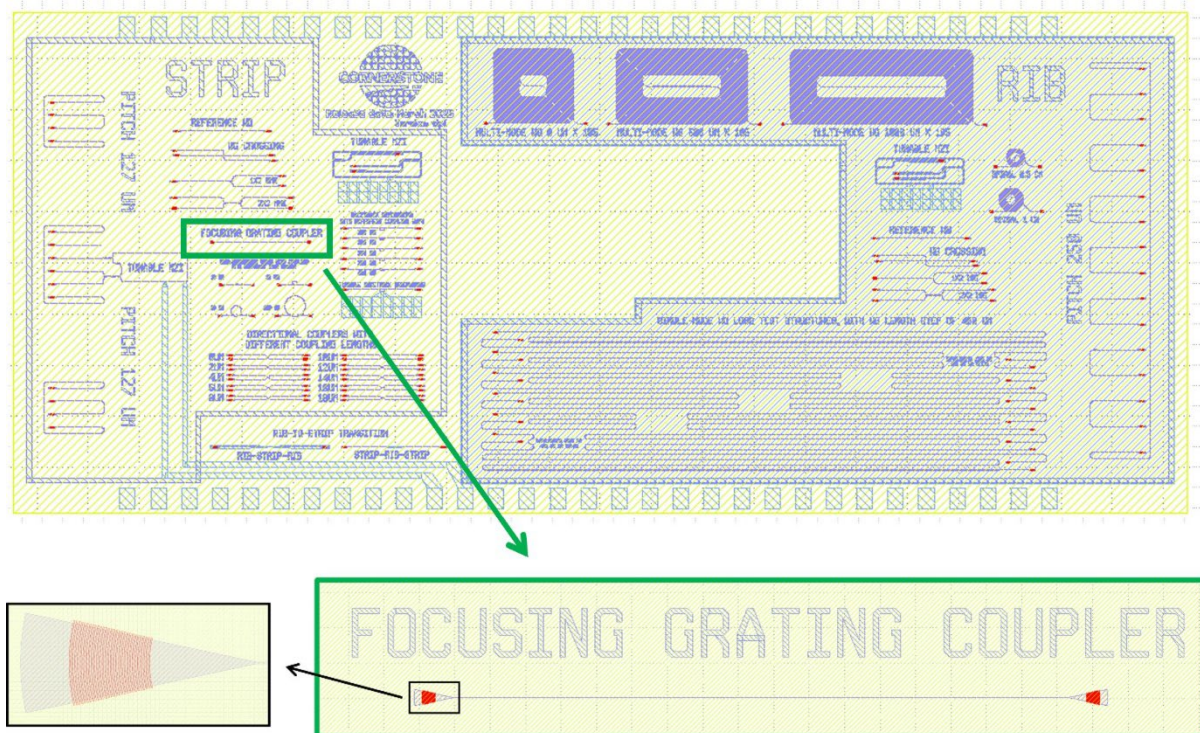


Figure 7. Focusing grating coupler

III. Photonic devices based on rib waveguides

There are also basic components and a tunable MZI on the rib side as marked in Fig. 8, which are very similar to the ones on the strip side. So they won't be discussed in detail. However, there is a point worth mentioning. The thermo-optic phase tuning efficiency in the rib MZI will be much lower than in the strip MZI because Si has a high thermal conductivity.

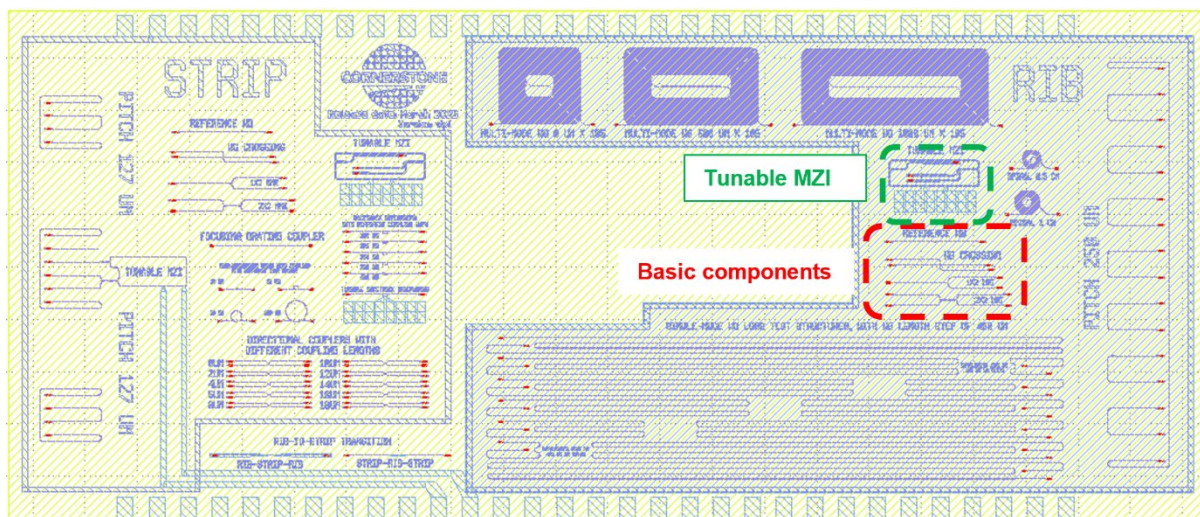


Figure 8. Basic components and tunable MZI on the rib side.

1. Waveguide loss test structures

As marked in Fig. 9, there are two sets of waveguide optical loss test structures for wavelengths of 1550 nm and 1310 nm, respectively. In each set, there are 6 waveguides with a waveguide length step of 400 μm , which is given in the physical mark on the chip. The wavelength and waveguide dimensions are also 哦呢 canin the

on-chip marks. If one measures the transmission in any set, and then plots and linearly fits the transmission in dB against the relative waveguide length, one can get the waveguide optical loss in dB per unit length. This is the so-called cutback measurement.

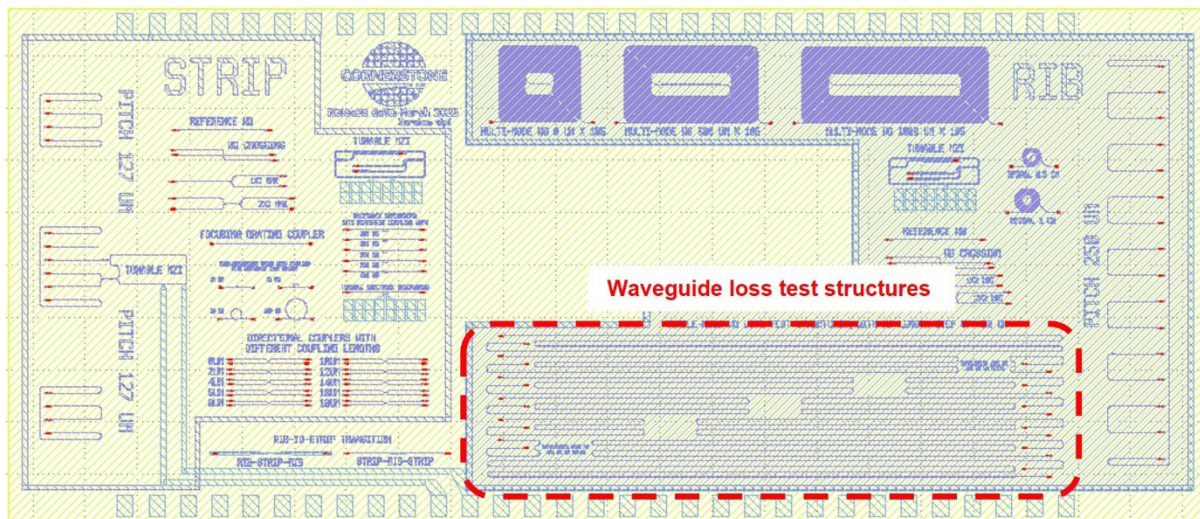


Figure 9. Waveguide loss test structures.

2. Spiral waveguides

As shown in Fig. 10, there are two types of spiral waveguide.

In Design A, the straight part of the waveguides are multi-mode (MM) waveguides, with a width of $2\ \mu\text{m}$, whilst the waveguide bend has to be SM to avoid mode crosstalk in the bend. They are connected using waveguide tapers as shown in the Fig. 10. There are three sets of spiral waveguides in Design A with stepped total waveguide length (optical path between the two gratings) and they can be used to measure the optical loss of the MM waveguide using the cutback method discussed above. Please note that the total waveguide length step is $500 \times 105\ \mu\text{m}$ as labelled on the chip.

In Design B, all the waveguides are arcs and thus SM waveguides. This is a popular design in sensing applications.

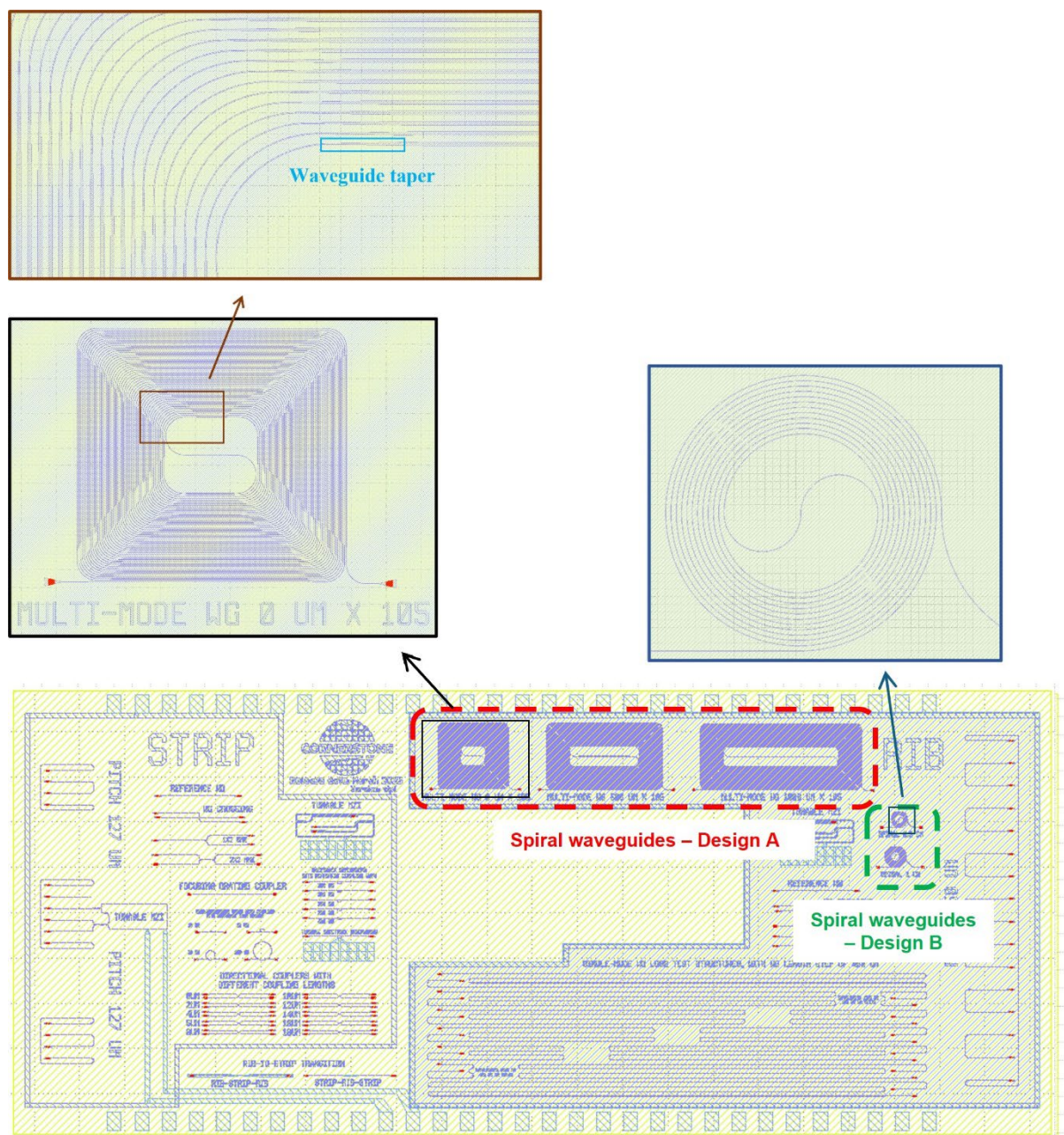


Figure 10. Spiral waveguides.

IV. Optical packaging – optical fibre bonding

For high-level integration, a PIC chip can be optically packaged. This PIC offers an example of optical packing: optical fibre bonding. As marked in Fig. 11, there are grating coupler arrays on both the strip side and the rib side.

On the strip side, the top and bottom arrays have four gratings and the middle one has six gratings. They all have a grating pitch of 127 μm . Note that in the middle device, a tunable MZI is connected to the gratings and the heater filament is connected to the contact pads on the bottom edge of the PIC, which can be wire bonded to a PCB. This device is used to demonstrate full PIC integration, including both optical packaging and electrical packaging.

On the rib side, there is only one array, which has 16 gratings with a pitch of 250 μm .

In all of these grating arrays, the first grating and the last one are always connected directly using a so-called 'optical shunt' to align the fibre array to the grating array during the fibre attach process, whilst the other gratings are functional gratings.

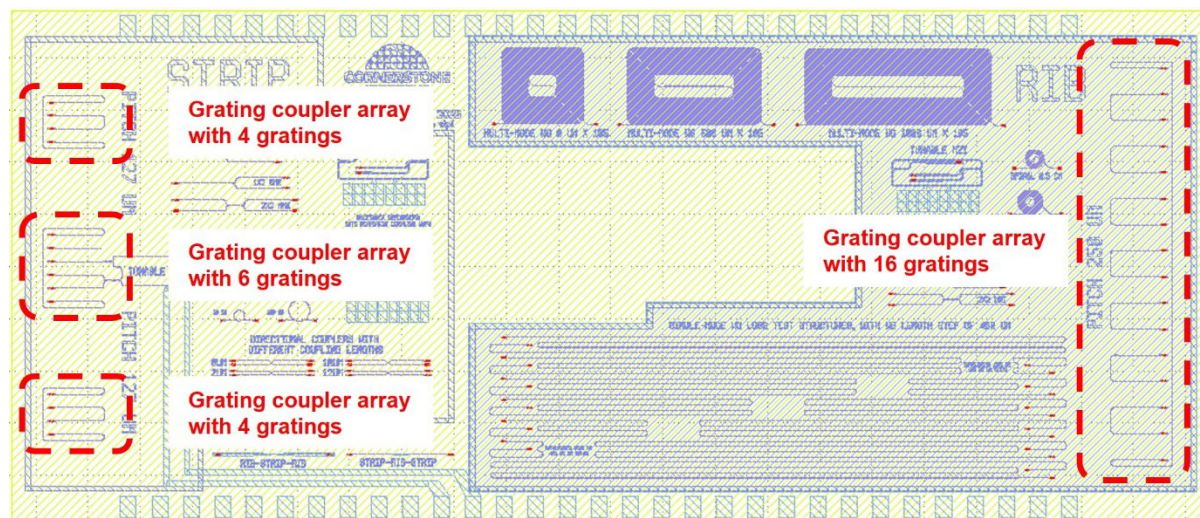


Figure 11. Grating coupler array template for optical fibre attachments.

V. Electrical packaging – wire bonding

For high-level integration, a PIC chip can be wire-bonded onto a PCB. As marked in Fig. 12, there are two rows of metal contact pads on the top edge and bottom edge of the PIC, which are the templates for the CORNERSTONE wire bonding service, which matches the pad layout on the standard CORNERSTONE PCBs.

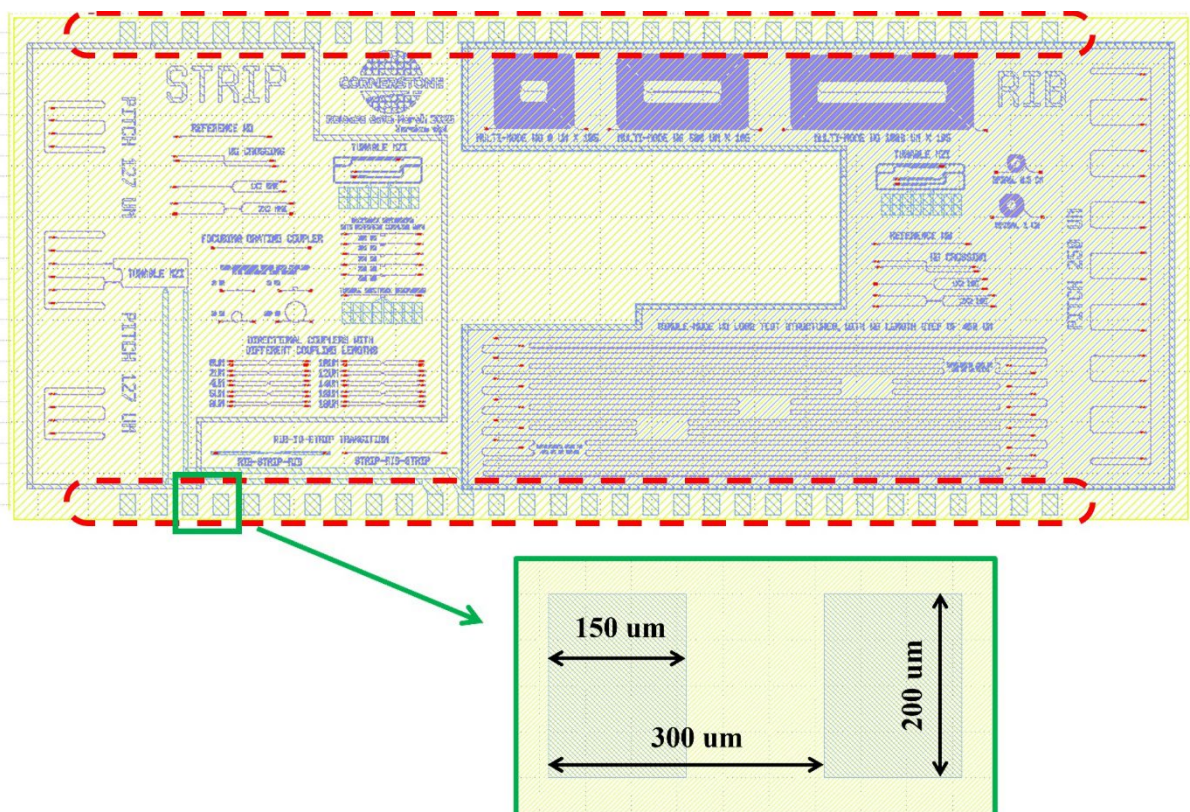


Figure 12. Metal contact pads template for wire bonding.